Sham Satish Thakare

Pre-Doctoral Researcher · Machine Learning, Distributed Systems & Computational Foundations

Capital One Systems Research · APCOER Pune (Savitribai Phule Pune University)

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RESEARCH INTERESTS

Foundations of Machine Learning & Reasoning: Reinforcement learning with verifiable rewards (RLVR), policy optimization (GRPO/PPO), uncertainty calibration, length confounding, and state-matched error recovery dynamics.
Fault-Tolerant Distributed Systems: State-machine replication (Raft, Paxos), dynamic quorum adaptation under network partitioning, linearizability verification, and tail latency minimization in modern C++20.
Mathematical & Quantum Foundations: Spectral graph theory, convex optimization, stochastic processes, quantum information foundations (density matrix formalism, unitary dynamics), and quantum circuit compilation complexity.

EDUCATION & HONORS

APCOER Pune (Savitribai Phule Pune University), Pune, India **01/2020 – 06/2024**

Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science | **Cumulative GPA: 8.70 / 10.00** **Rank 1 (9.53 SGPA)**

- **Gold Medal in Mathematics (2024):** Awarded for highest academic rank across mathematical and quantitative curricula in the university cohort.
- **Academic Standing:** Final Year SGPA **9.53 / 10.00** (First Class with Distinction, Department Rank 1).
- **Mathematics & Theory Coursework:** Advanced Linear Algebra (Spectral Decomposition, SVD), Probability & Stochastic Processes, Optimization Theory, Discrete Mathematics & Graph Theory, Calculus & Differential Equations.
- **Computer Science Coursework:** Design & Analysis of Algorithms, Data Structures, Distributed Computing, Operating Systems, Machine Learning, Deep Learning, Computational Intelligence, Database Management Systems.

RESEARCH & PROFESSIONAL EXPERIENCE

Capital One Systems Research, Pune, India **06/2024 – Present**

Pre-Doctoral Researcher (Distributed Systems & Machine Learning) | Mentorship: **Yuri Shelke** (Distinguished Engineer)

- Formulated dynamic quorum adaptation algorithms for Raft consensus clusters (quorumshift), proving invariant quorum overlap ($Q_i \cap Q_j \neq \emptyset$) under dynamic replica re-weighting without violating state-machine linearizability.
- Developed an online latency-profiling and anomaly-filtering estimator eliminating p99 tail latency degradation under asymmetric network fault injection and non-uniform round-trip delays.
- Engineered an asynchronous, lock-free C++20 distributed benchmarking harness evaluating multi-node consensus throughput across heterogeneous network partitions and emulated cross-datacenter WAN topologies.

APCOER Pune (Savitribai Phule Pune University), Pune, India **06/2023 – 05/2024**

Undergraduate Researcher & Teaching Assistant | Advisors: **Prof. Sagar Gawande** & **Prof. Sunil Thakare**

- Investigated token predictive entropy behavior in autoregressive reasoning models, discovering strong completion-length confounding ($r = +0.486$) that undermines heuristic uncertainty weighting in reinforcement learning.
- Designed state-matched continuation evaluation protocols isolating nominal error self-correction from unconditional generative capacity under frozen structural covariates.

• Served as Teaching Assistant for undergraduate courses in *Data Structures, Algorithms, and Foundations of Machine Learning*, conducting tutorials and recitations for cohorts of 120+ engineering students.

PUBLICATIONS & MANUSCRIPTS UNDER REVIEW

• When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in Language Model Reinforcement Learning

Sham Satish Thakare. *Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI)*, 2024. [Code] [Research Note]
Proved empirical completion-length confounding (r = +0.486, p < 10^-6) in token predictive entropy, proving that confidence weighting systematically penalizes valid multi-step derivations in GRPO; conducted five-way matched ablation study on GSM8K/MATH demonstrating 0.00% Pass@1 gain over length-controlled baselines.

• AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus

Sham Satish Thakare, **Yuri Shelke** (Distinguished Engineer, Capital One). *Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI)*, 2024. Capital One Systems Research. [Code (C++20)]
Formulated online latency profiling algorithms in C++20 Raft consensus clusters that dynamically re-weight quorums, proving non-empty intersection invariants ($Q_i \cap Q_j \neq \emptyset$) under bounded clock drift Δ, eliminating p99 tail latency degradation under asymmetric network jitter and partitions.

• recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

Sham Satish Thakare. *Under Review at 11th IEEE Special Session on Machine Learning on Big Data (IEEE BigData)*, 2024. [PDF Paper] [Code]
Formulated state-contrast continuation evaluation protocol ($D_{recovery} = E[\Delta_{recovery}] - E[\Delta_{control}]$) isolating true error recovery from baseline generative capacity under frozen structural covariates across reasoning model rollouts.

• MediRush-SafeAgent: Layered Runtime Enforcement for Healthcare-Commerce LLM Tool Execution

Sham Satish Thakare. *In Peer Review at Intelligence-Based Medicine (Elsevier) & SSRN Preprint (ID: 7446543)*, 2024. [SSRN] [Code]
Formulated multi-stage deterministic runtime policy interception and out-of-band schema validation barriers that decouple proposed calls from system mutations, guaranteeing 100% safe task completion and 0% policy violations.

PREPRINTS & WORKING PAPERS

• **StateShift: Tracking State-Dependent Reasoning Recovery Dynamics in Neural Sequence Models**: **Sham Satish Thakare**. *Technical Report & Preprint*, 2024. [PDF] [Code] — Evaluates continuation state matching under verifier perturbations and structural covariate controls.

• **Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?**: **Sham Satish Thakare**. *Working Paper in Preparation for Transactions on Machine Learning Research (TMLR)*, 2024. [Code & Benchmarks] — Characterizes compute Pareto frontiers comparing guided test-time search versus post-training RL.

• **Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints**: **Sham Satish Thakare**. *Technical Report & Diagnostic Probing Suite*, 2024. [Code] — Formulates gradient norm and feature drift diagnostic probes across model checkpoints, predicting reward plasticity prior to RL loops and reducing FLOPs by 18.7%.

ACADEMIC LEADERSHIP & SERVICE

• **President, Social Volunteer Club**, APCOER Pune (2023 – 2024): Elected student head of university community outreach; directed educational mentorship in quantitative and digital skills for 200+ underprivileged high-school students, organized city-wide blood donation drives, and led campus sustainability campaigns.

• **Teaching Assistant**, APCOER Pune (2023 – 2024): Conducted weekly recitation sessions, algorithmic proof walkthroughs (graph traversal, dynamic programming, complexity classes), and PyTorch implementation labs for 120+ undergraduate engineering students in *Data Structures, Algorithms, and Foundations of Machine Learning*.

TECHNICAL SKILLS & SCIENTIFIC TOOLS

• **Mathematics & Theoretical Foundations:** Linear Algebra (Spectral Decomposition, SVD), Probability Theory, Stochastic Processes, Convex Optimization (KKT Conditions, Subgradient Methods), Discrete Mathematics, Graph Theory, Algorithmic Complexity.

• **Quantum Computing & Information:** Quantum Gates & Circuits, Unitary Operators, Density Matrix Formalism, Quantum Algorithms (Shor, Grover, Quantum Walks), Qiskit, Cirq.

• **Machine Learning & AI:** Reinforcement Learning (PPO, GRPO, RLVR), Transformer Architectures, Statistical Learning Theory, PyTorch, JAX, Hugging Face Transformers, NumPy, SciPy.

• **Systems Programming & Infrastructure:** C++ (C++20 Multithreading, Lock-free Primitives), Distributed Consensus (Raft), POSIX, Linux Kernel Internals, OpenTelemetry, Git, Docker, LaTeX.

ACADEMIC & RESEARCH REFERENCES

• **Yuri Shelke**, Distinguished Engineer, Capital One · *Systems Research Mentor & Co-author (Distributed Systems & Consensus)* · Contact available upon request

• **Prof. Sagar Gawande**, Professor & Head of Department, APCOER Pune (SPPU) · *Undergraduate Research Advisor (AI & Machine Learning)* · Contact available upon request

• **Prof. Sunil Thakare**, Senior Faculty & Academic Mentor, APCOER Pune (SPPU) · *Mathematics & Foundations Mentor* · Contact available upon request